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Bottom block for direct chill casting

Abstract

A bottom block for use in direct chill (DC) casting that includes a flared base providing additional surface area for contact with coolant used in the DC casting process. The bottom block includes a first end that substantially fits within a DC mold while a second end has a larger cross-sectional area than the first end. A transition between the first end and the second end provides a surface for coolant to contact despite film boiling and air gaps between the ingot and the coolant.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 63/282,626, filed on Nov. 23, 2021, the entire contents of which is incorporated herein by reference for all purposes.

BACKGROUND

(1) Metal ingots are commonly produced by direct chill casting of molten metals. This involves pouring a molten metal into a mold having cooled walls, an open upper end and (after start-up) an open lower end. Molten metal is introduced into the mold at the open upper end and is cooled and

solidified (at least externally) as it passes through the mold. Solidified metal in the form of an ingot emerges from the open lower end of the mold and descends as the casting operation proceeds. Alternately, the casting takes place horizontally, but the procedure is essentially the same. Such casting techniques are particularly suited for the casting of aluminum and aluminum alloys but may be employed for other metals too.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The detailed description is set forth below with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items. The systems depicted in the accompanying figures are not to scale and components within the figures may be depicted not to scale with each other.
- (2) FIG. 1 illustrates a side view of a flared bottom block, according to at least one example.
- (3) FIG. 2 illustrates a section view of the flared bottom block of FIG. 1 taken across the width of the flared bottom block, according to at least one example.
- (4) FIG. 3 illustrates a top view of the flared bottom block of FIG. 1, according to at least one example.
- (5) FIG. 4 illustrates an end view of the flared bottom block of FIG. 1, according to at least one example.
- (6) FIG. 5 illustrates a side view of a flared bottom block, according to at least one example.
- (7) FIG. 6 illustrates a top view of the flared bottom block of FIG. 5, according to at least one example.
- (8) FIG. 7 illustrates an end view of the flared bottom block of FIG. 6, according to at least one example.
- (9) FIG. 8 illustrates a perspective view of a bottom block having ramped inner walls, according to at least one example.
- (10) FIG. 9 illustrates a perspective view of a bottom block having ramped inner walls, according to at least one example.
- (11) FIG. 10 illustrates a perspective view of a bottom block having removable segments for configurable lengths, according to at least one example.
- (12) FIG. 11 illustrates an end view of the bottom block of FIG. 10, according to at least one example.
- (13) FIG. 12 illustrates a side view of the bottom block of FIG. 10, according to at least one example.
- (14) FIG. 13 illustrates a perspective view of a bottom block having removable segments for configurable lengths, according to at least one example.
- (15) FIG. 14 illustrates a side view of the bottom block of FIG. 13, according to at least one example.
- (16) FIG. 15 illustrates a top view of the bottom block of FIG. 13, according to at least one example.
- (17) FIG. 16 illustrates a top view of a bottom block including a scalloped edge wall, according to at least one example.
- (18) FIG. 17 illustrates a perspective view of the bottom block of FIG. 16, according to at least one example.

DETAILED DESCRIPTION

- (19) Systems and devices described herein are related to an improved bottom block design particularly suitable for the vertical direct chill (DC) casting of large aluminum rolling ingots, i.e.,

having a cross section of at least 10×20 inches. The bottom blocks include an inner cavity such that when the bottom block is brought adjacent a DC mold body a temporary mold cavity is formed that may be filled to start the cast and the bottom block may be lowered once the cavity is filled with molten metal as is known in the art. The bottom portion of the ingot is sometimes referred to as the butt or butt portion of the ingot, and it tends to be thicker, which is sometimes referred to as “butt swell.” The examples described herein, which can be fabricated from aluminum or steel, may reduce butt swell and butt cracks and alligating in rolling ingots by providing thermal sinks and surface area for coolant to contact and further reduce the temperature of the bottom block and butt of the rolling ingot during casting. The example bottom blocks described herein enable contact with cooling fluid that typically bypasses the bottom block as it runs down the face of the ingot due to film boiling on the surface of the ingot that creates a barrier between the cooling liquid and the butt of the ingot and the bottom block, particularly as a DC cast progresses past an initial startup stage.

(20) In some examples, the term bottom block may also be referred to as a starting head, dummy block, stool cap, or a starting block, all commonly used to refer to the same general components.

(21) The bottom block designs described herein enable contact with cooling fluid, such as water, that is used in DC casting to cool a surface of the ingot. As mentioned above, due to film boiling at the surface of the ingot just after solidification, the cooling fluid typically does not run down the entire surface of the ingot and bottom block. As a result, in typical systems the bottom block becomes a source of retained heat that may keep the butt of the ingot at a higher temperature than desired and may cause increased butt swell or other adverse (distortion due to thermal differences in the block) effects. The designs herein enable additional surface contact with the cooling water as it flows downward during the cast and thereby reduces the heat stored in the bottom block, thereby reducing butt swell and other adverse effects.

(22) In some examples, a deeper bottom block may assist in reducing or eliminating bleed-outs of the cast ingot during the casting process so that the ingot may be able to support its own form and weight by the time the bottom block emerges below the mold walls. In some examples, the bottom block allows more time for the molten metal to remain and cool within the bottom block before additional weight from additional casting is placed on that initial metal, which allows that lower part of the solidifying ingot to better support the form of the ingot as a whole via solidification. In some examples, external or internal cooling within the bottom block may assist this support.

(23) The bottom block designs included herewith result in metal ingots that may be rolled following casting without introducing alligator splits at the end of the ingot by having a shaped end for the ingot that reduces the possibility of forming an alligator during rolling, thereby increasing metal yield during rolling. The shaped end formed by the bottom block tapers over the depth of the bottom block, which may be deeper than typical bottom blocks, and thereby reduce alligator shaped formations during rolling.

(24) The bottom block designs included herewith may reduce butt swell in ingots without requiring intentionally slowing of the cast speed, thereby improving the quality of, and maintaining speed of cast ingot output and reducing ingot processing to account for butt swell. The bottom block described herein promotes heat removal from the bottom block that may result in reduced butt swell, increased solidification rate, and improved metallographic qualities.

(25) The present description provides an overall understanding of the principles of the structure, function, manufacture, and use of the systems and methods disclosed herein. Several examples are illustrated in the accompanying drawings. Those of ordinary skill in the art will understand that the systems and methods specifically described herein and illustrated in the accompanying drawings are non-limiting embodiments. The features illustrated or described in connection with one embodiment may be combined with the features of other embodiments, including as between systems and methods. Such modifications and variations are intended to be included within the scope of the appended claims.

(26) Additional details are described below with reference to several example embodiments.

(27) FIGS. 1-4 illustrate a flared bottom block **100**, according to the instant disclosure. The flared bottom block **100** includes an upper end **104** configured to fit within an existing DC casting mold, and has a height, for example in a range from two centimeters to fifty centimeters, that enables engagement with a DC casting mold. The lower end **106** is larger in cross sectional area than the upper end **104** which results in sloped sides **102**. The sloped sides **102** provide a surface that extends outwards laterally from the sides of the bottom block **100**, beyond the dimensions of the ingot cast by the mold. Such protrusions enable cooling fluid, which would normally not contact the bottom block due to film boiling and other thermal effects or casting conditions, to contact the bottom block and run down the sloped sides **102**. The sloped sides **102** therefore provide a heat transfer surface for heat to be transferred from the bottom block **100** to the cooling fluid. In some examples, the sloped sides **102** may include or be replaced by other protrusions that extend beyond the cross section of the ingot and thereby contact cooling fluid. Such protrusions may include cooling fins, ridges, pyramids, and other such shapes or profiles that extend horizontally from the bottom block **100** and thereby increase a cross sectional surface area of a lower portion of the bottom block **100**.

(28) The bottom block **100** includes inner surfaces **208** that descend from the upper portion **104** towards a center of the bottom block, and thereby reduces the thickness of the ingot at the butt end that sits within the bottom block. The inner surfaces **208** provide for increased surface area contact with the butt of the cast ingot over typical bottom block designs and thereby increases heat transfer from the ingot to the bottom block **100**. The bottom block **100** may therefore become a heat sink that aids in removal of heat from the ingot and thereby reduces butt swell of ingots during casting. The additional contact with the cooling fluid aids in removal of heat from the bottom block **100** due to the flared sides and thereby further removes heat from the butt end of the ingot.

(29) In an embodiment, the bottom block **100** includes a ramped middle portion **314** and drain holes **316**. The ramped middle portion **314** may aid with distribution of molten metal during a starting phase of a DC cast, as metal may be initially introduced within the bottom block at a center of the bottom block **100** and flow down the ramped middle portions **314** towards the ends (along a long axis) of the bottom block **100**. The ramped middle portion **314** may also aid in positioning the butt of the ingot at the start of a cast along the centerline of the bottom block **100** so that the ingot does not shift later in the casting sequence and remains aligned to the center of the bottom block **100**. The drain holes **316** may enable coolant or other material to flow out of the bottom block **100** and prevent interference with the molten metal at the start of the cast.

(30) FIGS. 5-7 illustrate a flared bottom block **500**, according to the present disclosure. The flared bottom block **500** may include similar features to the bottom block **100** of FIGS. 1-4. For example, an upper end **504** may have a first cross-sectional area less than a lower end **506** with sloped sides **502** that connect between the upper end **504** and the lower end **506**. The flared bottom block **500** may also include a ramped middle portion **614** and drain holes **616** similar to the ramped middle portion **314** and drain holes **616** of FIGS. 1-4. The bottom block **500** may have a smaller height from the upper end **504** to the lower end **506** than the bottom block **100** and may be sized according to dimensions of existing bottom blocks to ensure compatibility with existing DC casting systems.

(31) FIGS. 8-9 illustrate a bottom block **800** having ramped inner walls **810**, according to the present disclosure. The bottom block **800** is fed, during DC casting, by a spout **802** and distribution device **804** that may direct molten metal from the spout **802** along a lengthwise axis of the bottom block **800** away from the spout **802**. The ramped middle portion **808**, which may be similar to the ramped middle portion **314** of FIGS. 1-4, may work in conjunction with the distribution device **804** to guide molten metal evenly around the bottom block **800**. The ramped middle portion **808** may also improve molten metal distribution within the bottom block **800** without inducing turbulent flow. The ramped middle portion **808** includes inclined surfaces that enhance laminar flow of metal

is the molten metal flows away from the spout **802** towards the edges of the bottom block **800**.

(32) The ramped inner walls **810** descend from the upper perimeter **806** towards a center of the bottom block, and thereby reduces the thickness of the ingot at the butt end that sits within the bottom block. The ramped inner walls may extend a distance of around one half the distance from a top of the bottom block to a bottom of the bottom block. In some examples, the ramped inner walls may extend more or less than half the distance from the top to the bottom of the bottom block, forming a deep cavity for receiving molten metal. The ramped inner walls **810** provide for increased surface area contact with the butt of the cast ingot and thereby increases heat transfer from the ingot to the bottom block **800**. The bottom block **800** may therefore become a heat sink that aids in removal of heat from the ingot and thereby reduces butt swell of ingots during casting.

(33) FIGS. **10-12** illustrate a bottom block **1000** having removable segments **1004** and **1010** for configurable lengths, according to at least one example. The flared bottom block **1000** may include similar features to the bottom block **100** of FIGS. **1-4** and/or other examples described herein. For example, an upper end **1018** may have a first cross-sectional area less than a lower end **1022** with sloped sides **1014** that connect between the upper end **1018** and the lower end **1014**. The upper end **1018** may be defined by perimeter walls **1020** that may interface with a mold body at the start of a casting procedure before being lowered out of the mold. The bottom block **1000** may also include drain holes, ramps, and other features similar to other examples described herein. The profile of the bottom block **1000** is designed with a shallow upper region that interfaces with a mold body, at the top end **1018**. The ends **1006** and **1012** also include ramped portions similar to the sloped sides **1014**.

(34) The removable segments **1004** and **1010** may be selectively removable to create varying widths of bottom blocks, for example to fit within particular mold sizes for rolling mills or other processing machinery that may require particular dimensions of metal products. The removable segments couple to middle portions **1002** and **1008** as well as to ends **1006** and **1012**. The middle portions **1002** and **1008** are shown forming a shallow cavity, though other types of bottom block designs may be implemented, such as described herein. The middle portions **1002** and **1008** may couple together using a fastener or other semi-permanent or removable attachment. In some examples, the middle portions **100** and **1008** may be permanently joined together.

(35) The removable segments **1004** and **1010** are similarly removable and insertable between the middle portions **1002** and **1008** and the ends **1006** and **1012**. Different numbers of removable sections may be used to adjust the width of the bottom block **1000**. In some examples, asymmetric bottom blocks may be designed by incorporating different numbers of removable sections **1004** and **1010**. In some examples, the removable segments **1004** and **1010** may have thicknesses of one inch or less, but in some examples may also have a thickness of greater than one inch, depending on a particular implementation. The removable segments **1004** and **1010** may be coupled together, with the bottom block **1000** assembled using fasteners such as bolts along the length of the bottom block **1000** to hold the removable segments **1010** to the middle portion **1008**, with the bolts running through the end **1012** to hold the end **1012** in place. The fasteners and/or pins or other alignment features may be used to ensure proper alignment of the segments for the bottom block **1000**.

(36) FIGS. **13-15** illustrate a bottom block **1300** having removable segments for configurable lengths, according to at least one example. The bottom block **1300** is shown with similar features to those described with respect to bottom block **1000**, including the middle portions **1302** and **1308**, removable segments **1304** and **1310**, ends **1306** and **1312**, top end **1318**, bottom end **1322**, and sloped sides **1314** and **1316**. The bottom block **1300** is shown with the perimeter walls **1320** having a height less than the height of the perimeter walls **1022**. The different heights between the bottom block **1000** and the bottom block **1300** may enable the bottom block **1000** to act as a heat sink for a larger capacity of heat energy due to the larger mass of the bottom block **1000**. However, the bottom block **1300** with the sloped sides **1314** and **1316** may also feature increased heat transfer to cooling fluid over typical bottom blocks and therefore may provide the benefits described herein.

(37) FIGS. **16-17** illustrate a bottom block **1600** including a scalloped perimeter wall, according to at least one example. The bottom block **1600** may be similar to the bottom block **100** and/or bottom block **800**, or other examples described herein. The bottom block **1600** therefore includes a top end **1604** defining a region area with a smaller cross-sectional area than a bottom end **1606**, with the external walls **1602** and internal walls **1608** sloped to taper from a narrower width at the top end **1604** to a greater width at the bottom end **1606**. The bottom block **1600** also includes ramps **1610** and drain holed **1612** similar to those described above.

(38) The external walls **1602** include cooling features configured to provide greater surface area for heat exchange with an environment and/or to a cooling fluid. In the illustrated example, the cooling features includes scalloped edges **1616** with raised portions **1614** between the scalloped edges **1616**. The scalloped edges **1616** provide for greater surface area and are shown having rounded and/or arced profiles, though other profiles are contemplated that would enable additional cooling through contact with fluids, such as a coolant as well as the surrounding air. The scalloped edges **1616** may have a conical or frusto-conical shape in some examples. For example, the cooling features may include fins or other such heat dissipating elements. In the illustrated example, the scalloped edges **1616** overcome a typical difficulty with adding such features to bottom blocks, longevity and strength requirements for the extreme temperatures and environments of casting. The scalloped edges may be machined to remove material in selected regions and thereby do not create delicate or fragile fins as typical heat exchangers would. Instead, the scalloped edges **1616** provide additional surface area as well as channeling of coolant into directed streams. In some examples, the scallops may direct the coolant to particular destinations, such as for circulation within the bottom block. In an example, the scalloped edges **1616** may funnel coolant into a passageway that passes through the bottom block before draining out the bottom end of the bottom block **1600**. In such examples, the bottom block may be integrally cooled without requiring separate coolant lines to be connected to the bottom block **1600**.

(39) The following describe examples that may be implemented, including examples that may be combined in various example embodiments.

(40) A. A bottom block for direct chill (DC) casting, comprising: a monolithic block defined by: an upper portion configured to interface with a DC casting mold; a lower portion having a cross-sectional area that is less than a cross sectional area defined by a perimeter of the upper portion; an external surface that tapers from the upper portion to the lower portion, the external surface having a slope that increases a width of the monolithic block as the external surface extends from the upper portion to the lower portion, wherein the external surface is configured to receive cooling fluid from a DC casting apparatus, the slope of the external surface configured to maintain the cooling fluid in contact along a height of the external surface; and an interior surface configured to receive molten metal for solidification.

(41) B. The bottom block of paragraph A, wherein the upper portion comprises a vertical surface to interface with the DC casting mold and wherein the external surface comprises a sloped portion adjacent the upper portion.

(42) C. The bottom block of paragraph A, wherein the external surface comprises one or more protrusions extending horizontally from the bottom block.

(43) D. The bottom block of paragraph A, wherein the interior surface comprises a central portion and a peripheral portion, the central portion at a first height and the peripheral portion at a second height, the first height greater than the second height.

(44) E. A bottom block for DC casting, comprising: an upper portion configured to interface with a DC casting mold; a lower portion configured to be supported by a DC casting apparatus for displacing the bottom block during casting; an inner wall having a first slope extending from the upper portion towards a center of the bottom block; and an inner surface that, together with the inner wall, defines a cavity of the bottom block for receiving molten metal, the inner surface comprising a central portion and a peripheral portion, the central portion at a first height and the

peripheral portion at a second height, the first height greater than the second height.

(45) F. The bottom block of paragraph E, wherein the inner wall extends at least half of a distance between the upper portion and the lower portion.

(46) G. The bottom block of paragraph E, further comprising an external wall comprising one or more protrusions extending away from a center of the bottom block and configured to interface with cooling liquid during casting to cool the bottom block.

(47) H. A bottom block for DC casting, comprising: an upper surface defining a receiving volume configured to receive molten metal during casting; a first end along a first direction; a second end along a second direction opposite the first direction; and one or more removable segments configured to change a length of the bottom block in response to the one or more removable segments being inserted or removed from the bottom block, the one or more removable segments configured to be inserted perpendicular to the first direction between the first end and the second end.

(48) I. The bottom block of paragraph H, wherein the bottom block comprises a first portion and a second portion, the first portion comprising the first end and the first portion and the second portion releasably secured together such that the one or more removable segments are insertable between the first portion and the second portion.

(49) J. The bottom block of paragraph H, further comprising one or more protrusions from an external surface of the bottom block configured to extend horizontally beyond a perimeter of the upper surface.

(50) K. The bottom block of paragraph H, wherein the one or more removable segments comprise a surface configured to interface with the upper surface to produce a continuous profile.

(51) L. A bottom block for DC casting, comprising: an upper portion configured to interface with a DC casting mold; a lower portion having a cross-sectional area that is less than a cross sectional area defined by a perimeter of the upper portion; an external surface that tapers from the upper portion to the lower portion, the external surface having a slope that increases a width of the bottom block as the external surface extends from the upper portion to the lower portion, wherein the external surface is configured to receive cooling fluid from a DC casting apparatus, the slope of the external surface configured to maintain the cooling fluid in contact along a height of the external surface, wherein the external surface includes one or more protrusions extending horizontally from the bottom block; and an interior surface configured to receive molten metal for solidification.

(52) M. The bottom block of paragraph L, wherein the one or more protrusions define channels extending at least a portion of a distance from the upper portion to the lower portion.

(53) N. The bottom block of paragraph L, wherein the protrusions comprise scalloped edges.

(54) O. The bottom block of paragraph L, wherein the interior surface comprises an inner wall having a first slope extending from the upper portion towards a center of the bottom block.

(55) While the foregoing invention is described with respect to the specific examples, it is to be understood that the scope of the invention is not limited to these specific examples. Since other modifications and changes varied to fit particular operating requirements and environments will be apparent to those skilled in the art, the invention is not considered limited to the example chosen for purposes of disclosure, and covers all changes and modifications which do not constitute departures from the true spirit and scope of this invention.

(56) Although the application describes embodiments having specific structural features and/or methodological acts, it is to be understood that the claims are not necessarily limited to the specific features or acts described. Rather, the specific features and acts are merely illustrative some embodiments that fall within the scope of the claims.

Claims

1. A bottom block for direct chill (DC) casting, comprising: a monolithic block defined by: an upper portion configured to interface with a DC casting mold; a lower portion having a cross-sectional area that is larger than a cross sectional area defined by a perimeter of the upper portion; an external surface that tapers from the upper portion to the lower portion, the external surface having a slope that increases a width of the monolithic block as the external surface extends from the upper portion to the lower portion, wherein the external surface is configured to receive cooling fluid from a DC casting apparatus, the slope of the external surface configured to maintain the cooling fluid in contact along a height of the external surface; and an interior surface configured to receive molten metal for solidification wherein the external surface comprises one or more protrusions extending horizontally from the bottom block.
 2. The bottom block of claim 1, wherein the upper portion comprises a vertical surface to interface with the DC casting mold and wherein the external surface comprises a sloped portion adjacent the upper portion.
 3. The bottom block of claim 1, wherein the interior surface comprises a central portion and a peripheral portion, the central portion at a first height and the peripheral portion at a second height, the first height greater than the second height.
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